Order Fulfilment & Installation Performance Analysis Dashboard

Business Operations Analytics Case Study · Neo Heat Pump – Sai Heating and Cooling Solutions

Orders Jan–Jun 2026 · Report as of 30 Jun 2026 · Prepared by [Your Name]

SIMULATED DATA: all 160 orders are synthetic records created for a portfolio case study. Results do not describe the real performance of the company. Recommendations are proposals, not implemented actions.

TOTAL ORDERS	ON-TIME DELIVERY	ON-TIME INSTALLATION	OVERDUE ORDERS
160	**72.4%**	**60.2%**	**9**
121 completed · 30 pending · 9 cancelled	37 late of 134	47 late of 118	7 stuck at installation

1. Business background

Neo Heat Pump (Sai Heating and Cooling Solutions) supplies heat pumps, heat-pump water heaters, swimming-pool heat pumps, heat-pump dryers and customer-specific solutions. Each order is manufactured, dispatched, delivered and then installed on the customer's site. That means two promises are made to the customer: a **delivery date** and an **installation date**. No company history, revenue or real performance figures are used in this case study.

2. Problem statement

Management has no single, trusted view of: how much work is completed, pending or cancelled; whether delivery and installation promises are kept; which open orders are already late; and which stage of the process causes the most delay.

3. Analytical objectives

1. Clean and validate the raw order export, with every change logged.
2. Define KPIs with explicit numerators, denominators and exclusions.
3. Build a filterable Excel dashboard and a web dashboard from the same rules.
4. Identify bottlenecks and recommend prioritised, measurable actions.

4. Data and methodology

Step	Detail
Data	163-row synthetic raw export → 160 unique orders; 5 product families, 3 customer types, 3 regions; anonymous customer codes; illustrative order values

Cleaning	3 duplicate rows removed; 24 inconsistent labels standardised; 6 alternate-format dates converted; 1 impossible date (31-02-2026) blanked and flagged; 3 impossible date sequences kept but excluded from timing; 48-entry cleaning log
Derived fields	Delivery/installation delay days, production/transit/installation durations, current stage, overdue flag and days overdue (judged at a fixed as-of date)
Tools	Excel (COUNTIFS, SUMIFS, AVERAGEIFS, INDEX/MATCH, LARGE, data validation, charts) · Python/pandas · Streamlit/Plotly
QA	Every workbook KPI recalculated independently in Python: 175 checks, 0 differences; status counts reconcile (121 + 30 + 9 = 160)

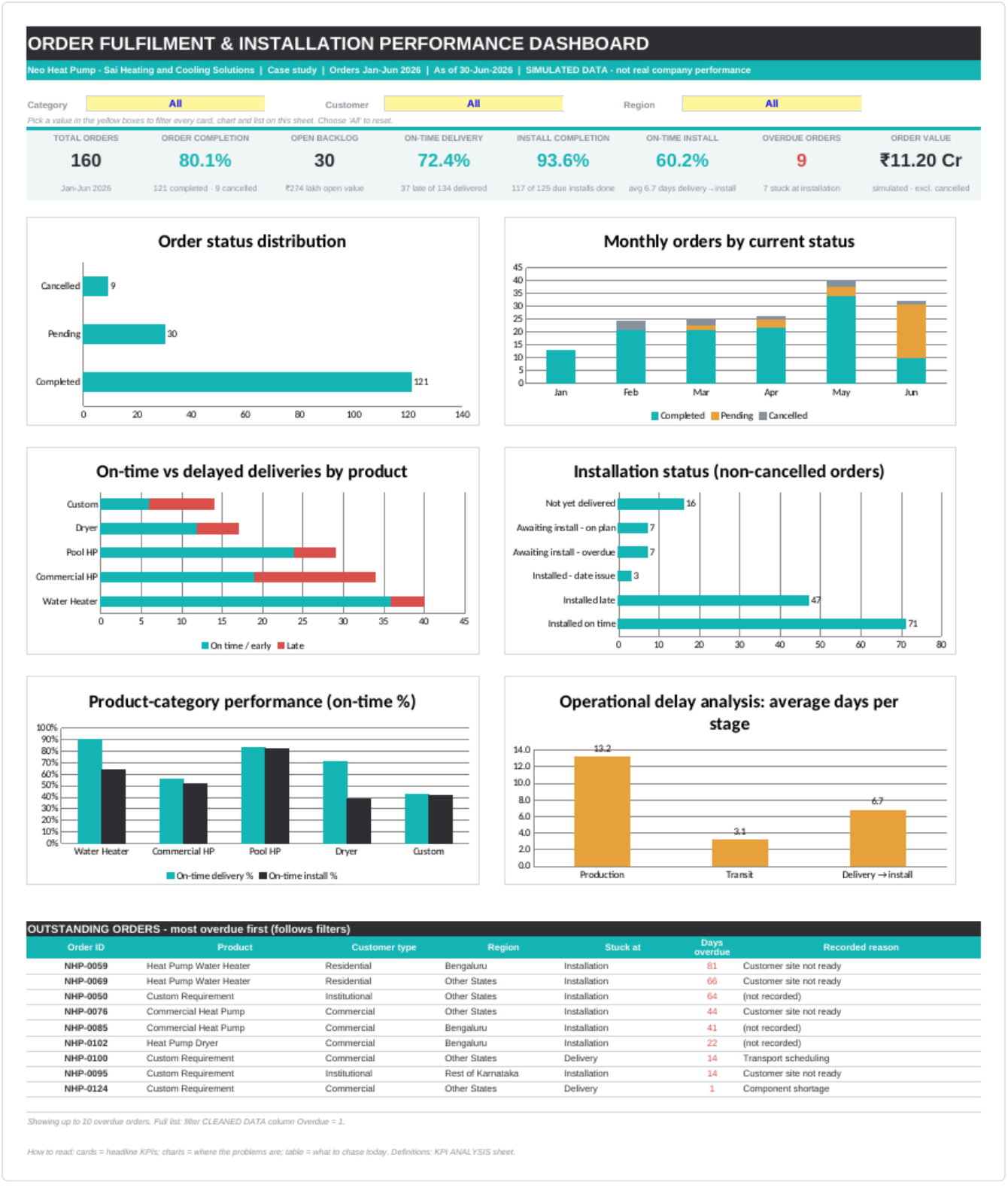
5. KPI calculations

KPI	Definition (numerator ÷ denominator)	Result
Order completion rate	Completed ÷ non-cancelled orders (121 ÷ 151)	**80.1%**
Cancellation rate	Cancelled ÷ total orders	5.6%
Current backlog	Pending orders (value, simulated)	30 (₹2.74 Cr)
Delivery completion rate	Due & delivered ÷ deliveries due (130 ÷ 132)	98.5%
On-time delivery rate	Delivered on/before promise ÷ delivered with valid dates (97 ÷ 134)	**72.4%**
Average delivery delay	Mean days late, late deliveries only (max 60)	6.7 d
Installation completion rate	Due & installed ÷ installations due (117 ÷ 125)	93.6%
On-time installation rate	Installed on/before promise ÷ installed with valid dates	**60.2%**
Pending installations	Delivered, not installed	14
Average installation duration	Delivery → installation	6.7 d
Overdue orders	Pending and past the promise for the current stage	**9** (avg 38.6 d)
Order cycle time	Order → installation, completed orders	22.4 d
Total order value	Non-cancelled, simulated	₹11.20 Cr

Cancelled orders are excluded from delivery and installation KPIs. Promises after the as-of date are "not yet due", never overdue.

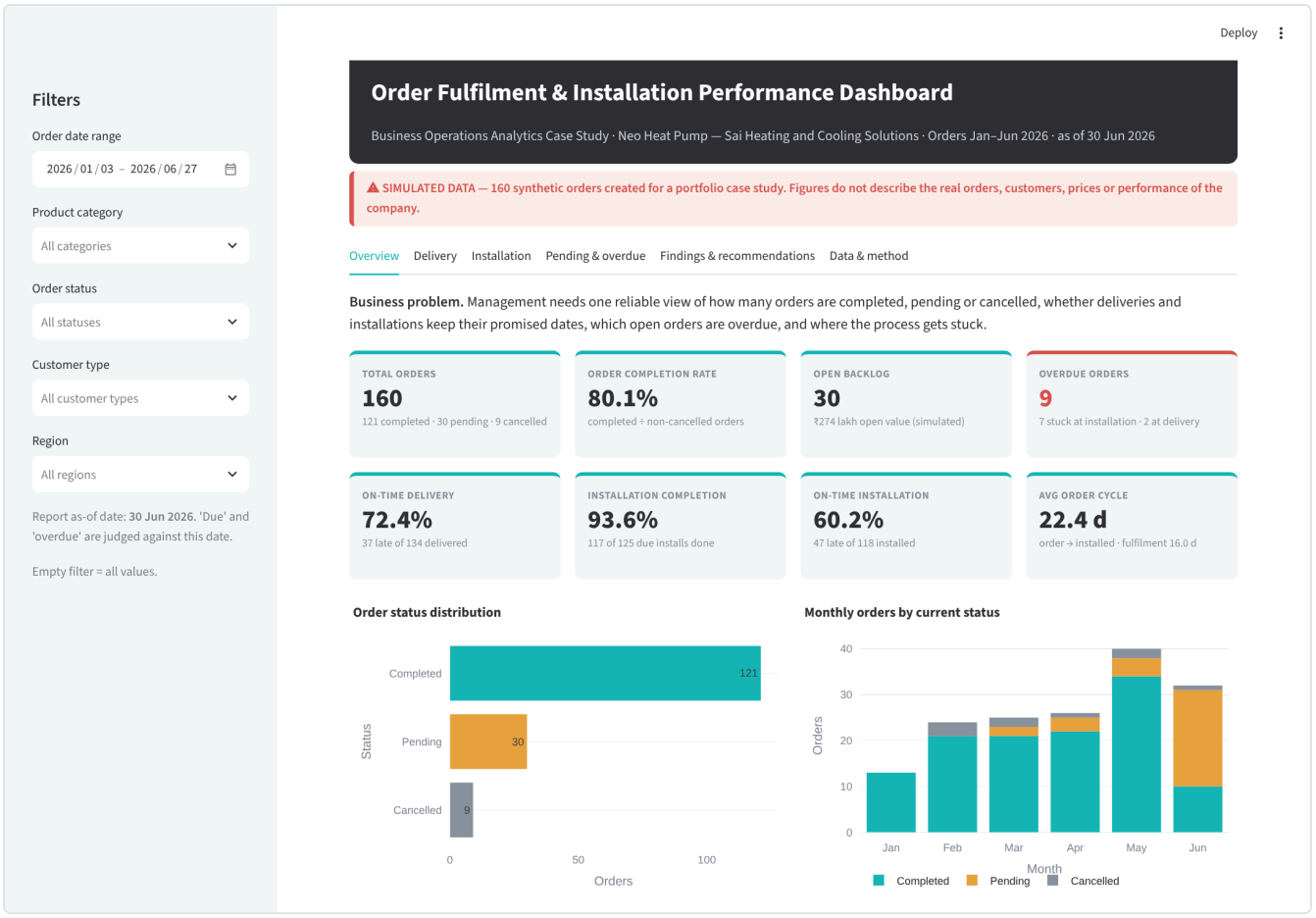
6. Dashboard design

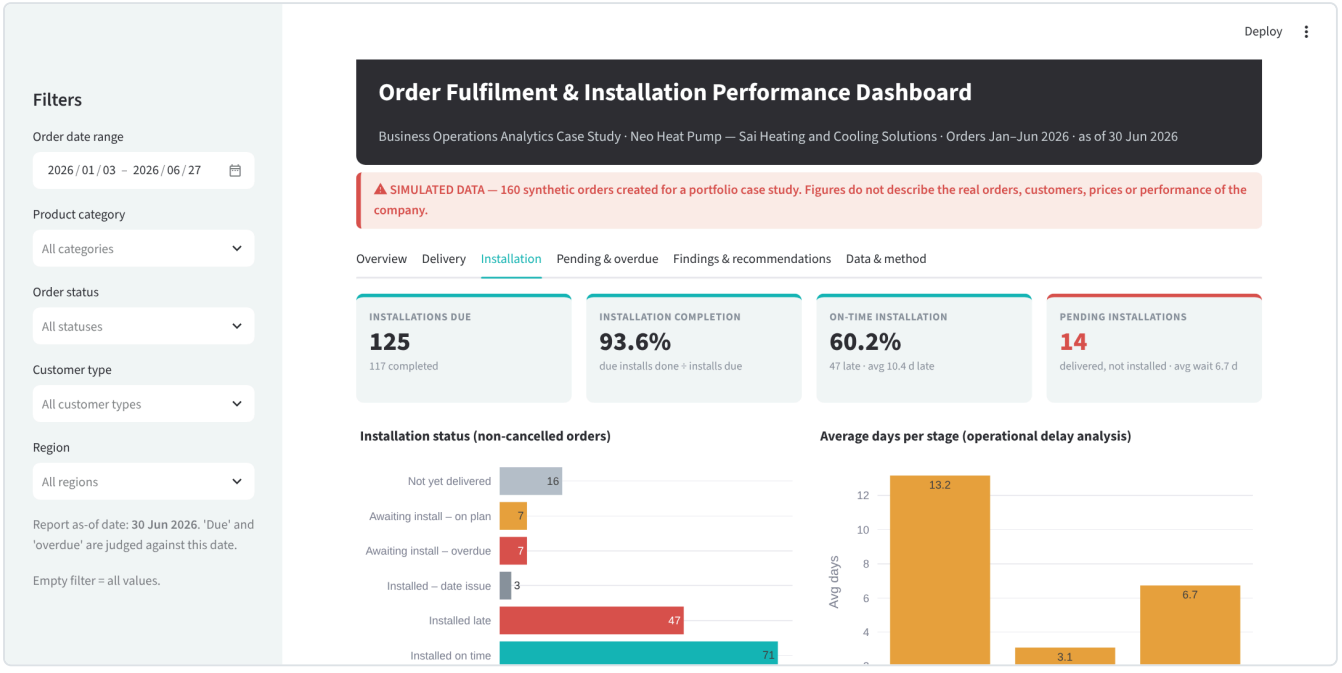
The Excel dashboard has 8 KPI cards, 6 charts and a "most overdue" action list. Three dropdown filters (category, customer type, region) drive every element through a helper column, so all values remain live formulas.



Web dashboard (Streamlit)

The web dashboard has six tabs (Overview, Delivery, Installation, Pending & overdue, Findings, Data & method) with filters for date, category, status, customer type and region. It reads the cleaned dataset and applies the same KPI rules.





7. Key findings

> **F1 · About 1 in 4 deliveries misses the promise.** On-time delivery is 72.4%, yet 98.5% of due deliveries were made. The problem is lateness, not missed deliveries.

F2 · Delays concentrate in Commercial and Custom orders. On-time delivery is 42.9% for Custom and 55.9% for Commercial, against 90.0% for water heaters. Together these two categories account for 23 of 37 late deliveries. Average production is 32.9 d for Custom against 5.0 d for water heaters.

F3 · Installation is the weakest stage. On-time installation is 60.2% against 72.4% for delivery, and installation completion is 93.6% against 98.5%.

F4 · Half of the late installations followed an on-time delivery. 24 of 47 did. The most frequent recorded reason was "Customer site not ready" (17 late installations).

F5 · Overdue work sits at installation. 9 orders are overdue: 7 at installation and 2 at delivery. They are 38.6 days overdue on average, and the oldest is 81 days.

F6 · Production is the largest share of cycle time. Production takes 13.2 d, transit 3.1 d and delivery→installation 6.7 d, against an average cycle of 22.4 d.

F7 · Other States are on time less often (52.4% against 78.5% for Bengaluru). This is a small sample of 21 timed deliveries.

F8 · Reasons are missing on 8 of 37 late deliveries and 14 of 47 late installations.

Causes are hypotheses supported only by recorded reason codes. Several patterns were assumptions built into the simulation.

8. Business recommendations

#	Recommendation	Impact / Effort / Urgency	Metric to monitor
R1	Daily review of overdue and delivered-not-installed orders	High / Low / Immediate	Overdue orders, days overdue
R2	Site-readiness check 3–5 days before delivery; book the installation slot only once the site is ready	High / Low / 2 weeks	On-time installation rate
R3	Promise dates by product category, based on actual production time	High / Medium / 1 month	On-time delivery by category
R4	Component check at order acceptance (Custom and Commercial)	Medium / Medium	Component-shortage delays
R5	Mandatory delay-reason dropdown	Medium / Low	% of late records with a reason
R6	One tracker with validated statuses and dates	Medium / Low	Data-quality flags per month
R7	Technician calendar and capacity plan	Medium / Medium	Delivery→installation days
R8	Weekly KPI report	Medium / Low	Report issued weekly
R9	Review transit allowance for Other States	Low / Low	On-time delivery, Other States

9. Limitations

- The data is simulated, so the results do not describe the company. Some patterns were designed into the generator.
- Groups are small: each product category has 14–40 timed deliveries, and Other States has only 21.
- Reason codes are self-reported and sometimes missing. Causes are hypotheses.
- There is no installation booking date, technician, cost or customer-satisfaction data.
- Overdue status depends on the chosen as-of date (30 Jun 2026).

10. What I learned

- Choosing the right **denominator** matters more than the chart. Completion, delivery completion, on-time delivery and installation completion are four different questions.
- Data cleaning is most of the work. Logging every change makes the numbers defensible.
- "Not yet due" is not "late". A fixed report date keeps the overdue logic honest.
- A finding is not a cause. Separating observations from hypotheses, and naming the data needed to confirm them, is part of the analyst's job.
- Reconciling results from two independent calculations catches mistakes before a manager does.

Repository: [GitHub URL] · Excel workbook: excel/operations_analysis.xlsx · Dashboard: dashboard/app.py